

$$\begin{aligned}
& \left(-\frac{1}{6} g_2^4 y_d^{i112} L F_{3,0}[m_2] - \frac{1}{6} g_2^4 y_d^{i112} L F_{4,-1}[m_2] + \frac{45}{8} g_2^4 y_d^{i112} L F_{5,-2}[m_2] + \frac{1}{12} g_2^2 y_d^{i112} \right. \\
& \quad \left. (-36 \overline{y_e^{pr}} y_d^{pr} + 5 \sum_p g_2^2) L F_{3,0}[m_P] - \frac{1}{48} g_2^4 y_d^{i112} (-24 \overline{y_e^{pr}} y_d^{pr} + \sum_p g_2^2) L F_{4,-1}[m_P] - \right. \\
& \quad \left. \frac{25}{45} \sum_p g_2^4 y_d^{i112} L F_{5,-2}[m_P] + \frac{1}{24} g_2^2 y_d^{i112} (-36 \overline{y_d^{pr}} y_d^{pr} + 5 \sum_p g_2^2) L F_{3,0}[m_q^P] - \right. \\
& \quad \left. \frac{1}{16} g_2^2 y_d^{i112} (-24 \overline{y_d^{pr}} y_d^{pr} + \sum_p g_2^2) L F_{4,-1}[m_q^P] - \frac{2}{15} \sum_p g_2^4 y_d^{i112} L F_{5,-2}[m_q^P] - \right. \\
& \quad \left. \frac{1}{18} g_2^4 y_d^{i112} L F_{3,0}[\tilde{\mu}] - \frac{1}{12} g_2^4 y_d^{i112} L F_{4,-1}[\tilde{\mu}] + \frac{45}{8} g_2^4 y_d^{i112} L F_{5,-2}[\tilde{\mu}] + \right. \\
& \quad \left. \frac{1}{9} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{2,1,0}[m_1, m_d^r] - \frac{2}{9} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{3,1,-1}[m_1, m_d^r] + \right. \\
& \quad \left. \frac{1}{9} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{4,1,-2}[m_1, m_d^r] - \frac{1}{108} g_1^4 y_d^{i112} L F_{2,2,-1}[m_1, m_d^{i2}] + \right. \\
& \quad \left. \frac{1}{36} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{2,1,0}[m_1, m_q^P] - \frac{1}{18} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{3,1,-1}[m_1, m_q^P] + \right. \\
& \quad \left. \frac{1}{36} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{4,1,-2}[m_1, m_q^P] - \frac{1}{864} g_1^2 y_d^{i112} (g_1^2 + 3 g_2^2) L F_{2,2,-1}[m_1, m_q^{i1}] + \right. \\
& \quad \left. \frac{1}{3} g_2^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{2,1,0}[m_2, m_q^P] - \frac{3}{2} g_2^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{3,1,-1}[m_2, m_q^P] + \right. \\
& \quad \left. \frac{1}{3} g_2^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{4,1,-2}[m_2, m_q^P] + \frac{1}{32} g_2^2 y_d^{i112} (-g_1^2 + g_2^2) L F_{2,2,-1}[m_2, m_q^{i1}] + \right. \\
& \quad \left. \frac{2}{3} g_2^4 y_d^{i112} L F_{2,1,0}[m_2, \tilde{\mu}] + \frac{17}{12} g_2^4 y_d^{i112} L F_{2,2,-1}[m_2, \tilde{\mu}] - \frac{1}{3} g_2^4 y_d^{i112} L F_{3,1,-1}[m_2, \tilde{\mu}] - \right. \\
& \quad \left. \frac{7}{3} g_2^4 y_d^{i112} L F_{3,2,-2}[m_2, \tilde{\mu}] + \frac{3}{2} g_2^4 y_d^{i112} L F_{3,2,-1}[m_2, \tilde{\mu}] + \right. \\
& \quad g_2^4 y_d^{i112} L F_{3,3,-3}[m_2, \tilde{\mu}] - g_2^4 m_1^2 y_d^{i112} L F_{3,3,-2}[m_2, \tilde{\mu}] + g_1^2 y_d^{i112} L F_{4,2,-3}[m_2, \tilde{\mu}] - \\
& \quad g_2^4 m_2^2 y_d^{i112} L F_{4,2,-2}[m_2, \tilde{\mu}] + \frac{4}{3} g_2^3 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{2,1,0}[m_3, m_d^r] - \\
& \quad \frac{8}{3} g_2^3 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{3,1,-1}[m_3, m_d^r] + \frac{4}{3} g_3^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{4,1,-2}[m_3, m_d^r] - \\
& \quad \frac{1}{9} g_1^2 g_3^2 y_d^{i112} L F_{2,2,-1}[m_3, m_d^{i2}] + \frac{4}{3} g_3^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{2,1,0}[m_3, m_q^P] - \\
& \quad \frac{8}{3} g_3^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{3,1,-1}[m_3, m_q^P] + \frac{4}{3} g_3^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{4,1,-2}[m_3, m_q^P] - \\
& \quad \frac{1}{18} g_3^2 y_d^{i112} (g_1^2 + 3 g_2^2) L F_{2,2,-1}[m_3, m_q^{i1}] - \frac{1}{4} g_1^2 y_d^{i112} \overline{a_d^{pr}} a_d^{pr} L F_{2,2,0}[m_d^r, m_q^P] + \\
& \quad \frac{1}{12} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{2,1,0}[m_d^r, \tilde{\mu}] - \frac{1}{24} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{2,2,-1}[m_d^r, \tilde{\mu}] + \\
& \quad \frac{1}{54} g_1^4 y_d^{i112} L F_{2,1,0}[m_d^{i2}, m_1] - \frac{2}{9} g_1^2 g_3^2 y_d^{i112} L F_{2,1,0}[m_d^{i2}, m_3] - \\
& \quad \frac{1}{4} g_2^2 y_d^{i112} \overline{a_e^{pr}} a_e^{pr} L F_{2,2,0}[m_e^r, m_P] - \frac{1}{2} g_2^2 y_d^{i112} \overline{a_e^{pr}} a_e^{pr} L F_{3,1,0}[m_P^P, m_e^r] + \\
& \quad \frac{1}{4} g_2^2 y_d^{i112} \overline{a_e^{pr}} a_e^{pr} L F_{3,2,-1}[m_P^P, m_e^r] + \frac{1}{8} y_d^{i112} \overline{a_e^{pr}} a_e^{pr} (3 g_1^2 + 5 g_2^2) L F_{4,1,-1}[m_P^P, m_e^r] - \\
& \quad \frac{1}{6} y_d^{i112} \overline{a_e^{pr}} a_e^{pr} (3 g_1^2 + g_2^2) L F_{5,1,-2}[m_P^P, m_e^r] + \\
& \quad \frac{1}{2} y_d^{i112} \overline{y_e^{pr}} y_e^{st} y_e^{pt} y_e^{sr} L F_{2,1,0}[m_P^P, m_e^s] - \frac{1}{2} y_d^{i112} \overline{y_e^{pr}} y_e^{st} y_e^{pt} y_e^{sr} L F_{3,1,-1}[m_P^P, m_e^s] - \\
& \quad \frac{1}{2} y_d^{i112} \overline{a_d^{pr}} a_d^{pr} (2 g_1^2 + 3 g_2^2) L F_{3,1,0}[m_P^P, m_d^r] + \frac{1}{4} g_1^2 y_d^{i112} \overline{a_d^{pr}} a_d^{pr} L F_{3,2,-1}[m_q^P, m_d^r] + \\
& \quad \frac{3}{6} y_d^{i112} \overline{a_d^{pr}} a_d^{pr} (7 g_1^2 + 5 g_2^2) L F_{4,1,-1}[m_q^P, m_d^r] - \\
& \quad \frac{1}{2} y_d^{i112} \overline{a_d^{pr}} a_d^{pr} (3 g_1^2 + g_2^2) L F_{5,1,-2}[m_q^P, m_d^r] + \\
& \quad \frac{3}{2} \overline{y_d^{pr}} y_d^{st} y_d^{pt} y_d^{sr} y_d^{i112} L F_{2,1,0}[m_q^P, m_q^s] - \frac{3}{2} \overline{y_d^{pr}} y_d^{st} y_d^{pt} y_d^{sr} y_d^{i112} L F_{3,1,-1}[m_q^P, m_q^s] - \\
& \quad \frac{1}{8} \tilde{\mu}^2 y_d^{i112} \overline{y_d^{pr}} y_d^{pr} (g_1^2 + g_2^2) L F_{2,2,0}[m_q^P, m_u^r] + \frac{3}{2} g_2^2 \tilde{\mu}^2 y_d^{i112} \overline{y_d^{pr}} y_d^{pr} L F_{3,1,0}[m_q^P, m_u^r] - \\
& \quad \frac{1}{2} g_2^2 \tilde{\mu}^2 y_d^{i112} \overline{y_d^{pr}} y_d^{pr} L F_{3,2,-1}[m_q^P, m_u^r] - \frac{3}{2} g_2^2 \tilde{\mu}^2 y_d^{i112} \overline{y_d^{pr}} y_d^{pr} L F_{4,1,-1}[m_q^P, m_u^r] + \\
& \quad \frac{1}{12} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{2,1,0}[m_q^P, \tilde{\mu}] - \frac{1}{24} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{2,2,-1}[m_q^P, \tilde{\mu}] - \\
& \quad \frac{1}{432} g_1^2 y_d^{i112} (g_1^2 + 3 g_2^2) L F_{2,1,0}[m_q^{i1}, m_3] + \frac{1}{4} \tilde{\mu}^2 y_d^{i112} \overline{y_d^{pr}} y_d^{pr} (g_1^2 - g_2^2) L F_{3,1,0}[m_u^r, m_q^P] + \\
& \quad \frac{1}{8} \tilde{\mu}^2 y_d^{i112} \overline{y_d^{pr}} y_d^{pr} (g_1^2 + g_2^2) L F_{3,2,-1}[m_u^r, m_r] + \frac{3}{4} \tilde{\mu}^2 y_d^{i112} \overline{y_d^{pr}} y_d^{pr} (g_1^2 + g_2^2) \\
& \quad L F_{4,1,-1}[m_u^r, m_q^P] - \frac{1}{2} \tilde{\mu}^2 y_d^{i112} \overline{y_d^{pr}} y_d^{pr} (3 g_1^2 + g_2^2) L F_{5,1,-2}[m_u^r, m_q^P] - \\
& \quad \frac{1}{6} g_1^2 y_d^{pi2} \overline{y_d^{pr}} y_d^{i1r} L F_{2,1,0}[m_u^r, \tilde{\mu}] + \frac{1}{12} g_1^2 y_d^{pi2} \overline{y_d^{pr}} y_d^{i1r} L F_{2,2,-1}[m_u^r, \tilde{\mu}] + \\
& \quad \frac{1}{2} g_1^2 y_d^{i112} (g_1^2 + g_2^2) L F_{3,1,-1}[\tilde{\mu}, m_1] - \frac{3}{8} g_1^4 y_d^{i112} L F_{3,2,-2}[\tilde{\mu}, m_1] - \\
& \quad \frac{3}{6} g_1^4 m_1^2 y_d^{i112} L F_{3,2,-1}[\tilde{\mu}, m_1] - \frac{1}{12} g_1^2 y_d^{i112} (12 g_1^2 + 7 g_2^2) L F_{4,1,-2}[\tilde{\mu}, m_1] + \\
& \quad \frac{1}{4} g_1^4 y_d^{i112} L F_{4,2,-3}[\tilde{\mu}, m_1] + \frac{1}{4} g_1^4 m_1^2 y_d^{i112} L F_{4,2,-2}[\tilde{\mu}, m_1] + \\
& \quad \frac{1}{6} g_1^2 y_d^{i112} (3 g_1^2 + g_2^2) L F_{5,1,-3}[\tilde{\mu}, m_1] + \frac{1}{6} g_2^2 y_d^{i112} (9 g_1^2 + g_2^2) L F_{3,1,-1}[\tilde{\mu}, m_2] - \\
& \quad \frac{85}{24} g_2^4 y_d^{i112} L F_{3,2,-2}[\tilde{\mu}, m_2] - \frac{3}{8} g_2^4 m_2^2 y_d^{i112} L F_{3,2,-1}[\tilde{\mu}, m_2] - \\
& \quad \frac{3}{4} g_2^2 y_d^{i112} (4 g_1^2 + g_2^2) L F_{4,1,-2}[\tilde{\mu}, m_2] + \frac{7}{4} g_2^4 y_d^{i112} L F_{4,2,-3}[\tilde{\mu}, m_2] + \\
& \quad \frac{1}{4} g_2^4 m_2^2 y_d^{i112} L F_{4,2,-2}[\tilde{\mu}, m_2] + \frac{1}{2} g_2^2 y_d^{i112} (3 g_1^2 + g_2^2) L F_{5,1,-3}[\tilde{\mu}, m_2] + \\
& \quad \frac{1}{2} \overline{y_d^{pr}} y_d^{st} y_d^{pi2} y_d^{sr} y_d^{i1t} L F_{2,1,0}[\tilde{\mu}, m_d^r] - \overline{y_d^{pr}} y_d^{st} y_d^{pi2} y_d^{sr} y_d^{i1t} L F_{3,1,-1}[\tilde{\mu}, m_d^r] + \\
& \quad \frac{1}{2} \overline{y_d^{pr}} y_d^{st} y_d^{pi2} y_d^{sr} y_d^{i1t} L F_{4,1,-2}[\tilde{\mu}, m_d^r] + \overline{y_d^{pr}} y_d^{st} y_d^{pi2} y_d^{sr} y_d^{i1t} L F_{2,1,0}[\tilde{\mu}, m_q^s] - \\
& \quad 2 \overline{y_d^{pr}} y_d^{st} y_d^{pi2} y_d^{sr} y_d^{i1t} L F_{3,1,-1}[\tilde{\mu}, m_q^s] + \overline{y_d^{pr}} y_d^{st} y_d^{pi2} y_d^{sr} y_d^{i1t} L F_{4,1,-2}[\tilde{\mu}, m_q^s] + \\
& \quad \frac{1}{2} \overline{y_d^{pr}} y_d^{si2} y_d^{i1r} y_d^{st} y_d^{pt} L F_{2,1,0}[\tilde{\mu}, m_u^t] - \overline{y_d^{pr}} y_d^{si2} y_d^{i1r} y_d^{st} y_d^{pt} L F_{3,1,-1}[\tilde{\mu}, m_u^t] + \\
& \quad \frac{1}{2} \overline{y_d^{pr}} y_d^{si2} y_d^{i1r} y_d^{st} y_d^{pt} L F_{4,1,-2}[\tilde{\mu}, m_u^t] + \frac{1}{18} m_1 g_1^2 y_d^{pi2} y_d^{i1r} \overline{a_d^{pr}} L F_{2,1,1,0}[m_1, m_d^r, m_q^P] - \\
& \quad \frac{1}{18} m_1 g_1^2 y_d^{pi2} y_d^{i1r} \overline{a_d^{pr}} L F_{3,1,1,-1}[m_1, m_d^r, m_q^P] + \frac{1}{36} m_1 g_1^2 \overline{y_d^{pr}} y_d^{pi2} a_d^{i1r} \\
& \quad L F_{2,1,1,0}[m_1, m_d^r, m_q^{i1}] + \frac{1}{72} m_1 g_1^2 \overline{y_d^{pr}} y_d^{pi2} a_d^{i1r} L F_{2,2,1,-1}[m_1, m_d^r, m_q^{i1}] - \\
& \quad \frac{1}{36} m_1 g_1^2 \overline{y_d^{pr}} y_d^{pi2} a_d^{i1r} L F_{3,1,1,-1}[m_1, m_d^r, m_q^{i1}] + \frac{1}{3} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{1,1,1,0}[m_1, m_d^r, \tilde{\mu}] - \\
& \quad \frac{2}{3} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{2,1,1,-1}[m_1, m_d^r, \tilde{\mu}] + \frac{1}{3} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{3,1,1,-2}[m_1, m_d^r, \tilde{\mu}] - \\
& \quad \frac{1}{9} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{1,1,1,0}[m_1, m_d^{i2}, m_d^r] + \frac{1}{18} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{2,1,1,-1}[m_1, m_d^{i2}, m_d^r] + \\
& \quad \frac{1}{36} m_1 g_1^2 \overline{y_d^{pr}} y_d^{i1r} a_d^{pi2} L F_{2,1,1,0}[m_1, m_d^{i2}, m_q^P] - \\
& \quad \frac{1}{72} m_1 g_1^2 \overline{y_d^{pr}} y_d^{i1r} a_d^{pi2} L F_{2,2,1,-1}[m_1, m_d^{i2}, m_q^P] - \frac{1}{36} m_1 g_1^2 \overline{y_d^{pr}} y_d^{i1r} a_d^{pi2} \\
& \quad L F_{3,1,1,-1}[m_1, m_d^{i2}, m_q^P] - \frac{1}{144} m_1 g_1^2 a_d^{i112} (g_1^2 + g_2^2) L F_{2,2,1,-1}[m_1, m_d^{i2}, m_q^{i1}] - \\
& \quad \frac{1}{12} g_1^4 y_d^{i112} L F_{2,1,1,-1}[m_1, m_d^{i2}, \tilde{\mu}] - \frac{1}{18} g_1^4 m_1^2 y_d^{i112} L F_{2,1,1,0}[m_1, m_d^{i2}, \tilde{\mu}] + \\
& \quad \frac{1}{18} g_1^4 y_d^{i112} L F_{3,1,1,-2}[m_1, m_d^{i2}, \tilde{\mu}] + \frac{1}{18} g_1^4 m_1^2 y_d^{i112} L F_{3,1,1,-1}[m_1, m_d^{i2}, \tilde{\mu}] + \\
& \quad \frac{1}{6} m_1 g_1^2 \overline{y_d^{pr}} y_d^{i1r} a_d^{pi2} L F_{2,2,1,-1}[m_1, m_q^{i2}, m_q^P] + \frac{1}{6} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{1,1,1,0}[m_1, m_q^P, \tilde{\mu}] - \\
& \quad \frac{1}{6} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{2,1,1,-1}[m_1, m_q^P, \tilde{\mu}] + \frac{1}{6} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{3,1,1,-2}[m_1, m_q^P, \tilde{\mu}] - \\
& \quad \frac{1}{72} m_1 g_1^2 \overline{y_d^{pr}} y_d^{pi2} a_d^{i1r} L F_{2,2,1,-1}[m_1, m_q^{i1}, m_d^r] - \frac{1}{144} m_1 g_1^2 a_d^{i112} (g_1^2 + g_2^2) \\
& \quad L F_{2,2,1,-1}[m_1, m_q^{i1}, m_d^{i2}] - \frac{1}{36} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{1,1,1,0}[m_1, m_q^{i1}, m_q^P] + \\
& \quad \frac{1}{72} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{2,1,1,-1}[m_1, m_q^{i1}, m_q^P] - \frac{1}{48} g_1^4 y_d^{i112} L F_{2,1,1,-1}[m_1, m_q^{i1}, \tilde{\mu}] - \\
& \quad \frac{1}{72} g_1^4 m_1^2 y_d^{i112} L F_{2,1,1,0}[m_1, m_q^{i1}, \tilde{\mu}] + \frac{1}{72} g_1^4 y_d^{i112} L F_{3,1,1,-2}[m_1, m_q^{i1}, \tilde{\mu}] + \\
& \quad \frac{1}{72} g_1^4 m_1^2 y_d^{i112} L F_{3,1,1,-1}[m_1, m_q^{i1}, \tilde{\mu}] + \frac{3}{8} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{2,2,1,-2}[m_1, \tilde{\mu}, m_d^r] + \\
& \quad \frac{1}{72} g_1^2 (6 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} + y_d^{i112} (11 g_1^2 - 3 g_2^2)) L F_{2,2,1,-2}[m_1, \tilde{\mu}, m_q^{i2}] + \\
& \quad \frac{1}{36} g_1^4 m_1^2 y_d^{i112} L F_{2,2,1,-1}[m_1, \tilde{\mu}, m_d^{i2}] + \frac{1}{4} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{2,2,1,-2}[m_1, \tilde{\mu}, m_q^P] + \\
& \quad \frac{1}{144} g_1^2 (6 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} + y_d^{i112} (10 g_1^2 - 3 g_2^2)) L F_{2,2,1,-2}[m_1, \tilde{\mu}, m_q^{i1}] + \\
& \quad \frac{13}{144} g_1^4 m_1^2 y_d^{i112} L F_{2,2,1,-1}[m_1, \tilde{\mu}, m_q^{i1}] + \frac{3}{2} g_2^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{1,1,1,0}[m_2, m_q^P, \tilde{\mu}] - \\
& \quad 3 g_2^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{2,1,1,-1}[m_2, m_q^P, \tilde{\mu}] + \frac{3}{2} g_2^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{3,1,1,-2}[m_2, m_q^P, \tilde{\mu}] - \\
& \quad \frac{1}{4} g_2^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{1,1,1,0}[m_2, m_q^{i1}, m_q^P] + \frac{1}{8} g_2^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{2,1,1,-1}[m_2, m_q^{i1}, m_q^P] - \\
& \quad \frac{15}{16} g_2^4 y_d^{i112} L F_{2,1,1,-1}[m_2, m_q^{i1}, \tilde{\mu}] - \frac{1}{8} g_2^4 m_2^2 y_d^{i112} L F_{2,1,1,0}[m_2, m_q^{i1}, \tilde{\mu}] + \\
& \quad \frac{5}{8} g_2^4 y_d^{i112} L F_{3,1,1,-2}[m_2, m_q^{i1}, \tilde{\mu}] + \frac{1}{8} g_2^4 m_2^2 y_d^{i112} L F_{3,1,1,-1}[m_2, m_q^{i1}, \tilde{\mu}] - \\
& \quad 2 g_1^2 g_2^2 y_d^{i112} L F_{2,2,1,-2}[m_2, \tilde{\mu}, m_1] - \frac{3}{2} m_1 m_2 g_1^2 g_2^2 y_d^{i112} L F_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] + \\
& \quad g_1^2 g_2^2 y_d^{i112} L F_{3,2,1,-3}[m_2, \tilde{\mu}, m_1] + m_1 m_2 g_1^2 g_2^2 y_d^{i112} L F_{3,2,1,-2}[m_2, \tilde{\mu}, m_1] + \\
& \quad \frac{1}{8} g_2^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{2,2,1,-2}[m_2, \tilde{\mu}, m_d^r] + \frac{3}{2} g_2^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{2,2,1,-2}[m_2, \tilde{\mu}, m_q^P] + \\
& \quad \frac{1}{16} g_2^2 (6 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} + y_d^{i112} (-3 g_1^2 + 22 g_2^2)) L F_{2,2,1,-2}[m_2, \tilde{\mu}, m_q^{i1}] + \\
& \quad \frac{1}{16} g_2^4 m_2^2 y_d^{i112} L F_{2,2,1,-1}[m_2, \tilde{\mu}, m_q^{i1}] + \frac{1}{4} g_2^2 \tilde{\mu}^2 y_d^{pi2} \overline{y_d^{pr}} y_d^{i1r} L F_{2,2,1,-1}[m_2, \tilde{\mu}, m_u^r] - \\
& \quad \frac{4}{3} m_3 g_3^2 y_d^{pi2} y_d^{i1r} \overline{a_d^{pr}} L F_{2,1,1,0}[m_3, m_d^r, m_q^P] + \\
& \quad \frac{4}{3} m_3 g_3^2 y_d^{pi2} y_d^{i1r} \overline{a_d^{pr}} L F_{3,1,1,-1}[m_3, m_d^r, m_q^P] - \\
& \quad \frac{2}{3} m_3 g_3^2 \overline{y_d^{pr}} y_d^{pi2} a_d^{i1r} L F_{2,1,1,0}[m_3, m_d^r, m_q^{i1}] - \frac{1}{3} m_3 g_3^2 \overline{y_d^{pr}} y_d^{pi2} a_d^{i1r} \\
& \quad L F_{2,2,1,-1}[m_3, m_d^r, m_q^{i1}] + \frac{2}{3} m_3 g_3^2 \overline{y_d^{pr}} y_d^{pi2} a_d^{i1r} L F_{3,1,1,-1}[m_3, m_d^r, m_q^{i1}] - \\
& \quad \frac{4}{3} g_3^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{1,1,1,0}[m_3, m_d^{i2}, m_d^r] + \frac{2}{3} g_3^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{2,1,1,-1}[m_3, m_d^{i2}, m_d^r] - \\
& \quad \frac{2}{3} m_3 g_3^2 \overline{y_d^{pr}} y_d^{i1r} a_d^{pi2} L F_{2,1,1,0}[m_3, m_d^{i2}, m_q^P] + \frac{2}{3} m_3 g_3^2 \overline{y_d^{pr}} y_d^{i1r} a_d^{pi2} \\
& \quad L F_{3,1,1,-1}[m_3, m_d^{i2}, m_q^P] + \frac{1}{6} m_3 g_3^2 a_d^{i112} (g_1^2 + g_2^2) L F_{2,2,1,-1}[m_3, m_d^{i2}, m_q^{i1}] - \\
& \quad \frac{1}{3} m_3 g_3^2 \overline{y_d^{pr}} y_d^{i1r} a_d^{pi2} L F_{2,2,1,-1}[m_3, m_q^P, m_d^{i2}] + \frac{1}{3} m_3 g_3^2 \overline{y_d^{pr}} y_d^{pi2} a_d^{i1r} \\
& \quad L F_{2,2,1,-1}[m_3, m_q^{i1}, m_d^r] + \frac{1}{6} m_3 g_3^2 a_d^{i112} (g_1^2 + g_2^2) L F_{2,2,1,-1}[m_3, m_q^{i1}, m_d^{i2}] - \\
& \quad \frac{4}{3} g_3^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{1,1,1,0}[m_3, m_q^{i1}, m_q^P] + \frac{2}{3} g_3^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{2,1,1,-1}[m_3, m_q^{i1}, m_q^P] - \\
& \quad \frac{1}{18} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{2,1,1,-1}[m_d^r, m_1, m_d^{i2}] - \\
& \quad \frac{2}{3} g_3^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{2,1,1,-1}[m_d^r, m_3, m_d^{i2}] - \frac{1}{2} \overline{y_d^{pr}} y_d^{st} y_d^{pi2} y_d^{sr} y_d^{i1t} \\
& \quad L F_{1,1,1,0}[m_d^r, m_q^t, \tilde{\mu}] - \frac{1}{4} \overline{y_d^{pr}} y_d^{st} y_d^{pi2} y_d^{sr} y_d^{i1t} L F_{2,1,1,-1}[m_d^r, m_q^t, \tilde{\mu}] + \\
& \quad \frac{3}{4} \overline{y_d^{st}} y_d^{sr} y_d^{i112} \overline{a_d^{pr}} a_d^{pt} L F_{2,2,1,-1}[m_d^r, m_q^P, m_q^t] + \frac{1}{4} \overline{y_d^{pr}} y_d^{st} y_d^{pi2} y_d^{sr} y_d^{i1t} \\
& \quad L F_{2,1,1,-1}[m_d^t, m_d^r, \tilde{\mu}] - \frac{3}{4} \overline{y_d^{st}} y_d^{sr} y_d^{i112} \overline{a_d^{pr}} a_d^{pt} L F_{2,2,1,-1}[m_d^t, m_q^P, m_d^r] + \\
& \quad \frac{1}{18} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{2,1,1,-1}[m_d^{i2}, m_1, m_d^r] + \frac{1}{216} m_1 g_1^2 a_d^{i112} (7 g_1^2 + 3 g_2^2) \\
& \quad L F_{2,1,1,0}[m_d^{i2}, m_1, m_q^{i1}] - \frac{1}{72} m_1 g_1^2 a_d^{i112} (g_1^2 + g_2^2) L F_{3,1,1,-1}[m_d^{i2}, m_1, m_q^{i1}] - \\
& \quad \frac{1}{18} g_1^4 y_d^{i112} L F_{2,1,1,-1}[m_d^{i2}, m_1, \tilde{\mu}] + \frac{2}{3} g_3^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{2,1,1,-1}[m_d^{i2}, m_3, m_d^r] - \\
& \quad \frac{1}{9} m_3 g_3^2 a_d^{i112} (7 g_1^2 + 3 g_2^2) L F_{2,1,1,0}[m_d^{i2}, m_3, m_q^{i1}] + \\
& \quad \frac{1}{3} m_3 g_3^2 a_d^{i112} (g_1^2 + g_2^2) L F_{3,1,1,-1}[m_d^{i2}, m_3, m_q^{i1}] + \\
& \quad \frac{1}{4} y_d^{i112} \overline{y_e^{st}} y_e^{sr} \overline{a_e^{pr}} a_e^{pt} L F_{2,2,1,-1}[m_e^r, m_P^P, m_e^t] - \\
& \quad \frac{1}{4} y_d^{i112} \overline{y_e^{st}} y_e^{sr} \overline{a_e^{pr}} a_e^{pt} L F_{2,2,1,-1}[m_e^t, m_P^P, m_e^r] + \\
& \quad \frac{1}{2} y_d^{i112} \overline{y_e^{st}} y_e^{sr} \overline{a_e^{pr}} a_e^{pt} L F_{2,1,1,0}[m_P^P, m_e^r, m_e^t] - \\
& \quad \frac{1}{2} y_d^{i112} \overline{y_e^{st}} y_e^{sr} \overline{a_e^{pr}} a_e^{pt} L F_{3,1,1,-1}[m_P^P, m_e^r, m_e^t] + \\
& \quad y_d^{i112} \overline{y_e^{st}} y_e^{pt} \overline{a_e^{pr}} a_e^{sr} L F_{2,1,1,0}[m_P^P, m_e^r, m_e^s] - y_d^{i112} \overline{y_e^{st}} y_e^{pt} \overline{a_e^{pr}} a_e^{sr} \\
& \quad L F_{3,1,1,-1}[m_P^P, m_e^r, m_e^s] - \frac{1}{72} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{2,1,1,-1}[m_q^P, m_1, m_q^{i1}] - \\
& \quad \frac{1}{6} g_2^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{2,1,1,-1}[m_q^P, m_2, m_q^{i1}] - \frac{2}{3} g_3^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} L F_{2,1,1,-1}[m_q^P, m_3, m_q^{i1}] + \\
& \quad \frac{3}{2} \overline{y_d^{st}} y_d^{sr} y_d^{i112} \overline{a_d^{pr}} a_d^{pt} L F_{2,1,1,0}[m_q^P, m_d^r, m_d^t] - \\
& \quad \frac{3}{2} \overline{y_d^{st}} y_d^{sr} y_d^{i112} \overline{a_d^{pr}} a_d^{pt} L F_{3,1,1,-1}[m_q^P, m_d^r, m_d^t] + \\
& \quad 3 \overline{y_d^{st}} y_d^{pt} y_d^{i112} \overline{a_d^{pr}} a_d^{sr} L F_{2,1,1,0}[m_q^P, m_d^r, m_q^s] - \\
& \quad 3 \overline{y_d^{st}} y_d^{pt} y_d^{i112} \overline{a_d^{pr}} a_d^{sr} L F_{3,1,1,-1}[m_q^P, m_d^r, m_q^s] - \\
& \quad \frac{3}{2} \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{sr} y_d^{i112} \overline{y_d^{st}} y_d^{pt} L F_{2,1,1,0}[m_q^P, m_q^s, m_u^t] + \\
& \quad \frac{3}{2} \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{sr} y_d^{i112} \overline{y_d^{st}} y_d^{pt} L F_{3,1,1,-1}[m_q^P, m_q^s, m_u^t] - \\
& \quad \frac{1}{2} \overline{y_d^{pr}} y_d^{st} y_d^{pt} y_d^{si2} y_d^{i1r} L F_{1,1,1,0}[m_q^P, m_q^s, \tilde{\mu}] + \\
& \quad \frac{1}{4} \overline{y_d^{pr}} y_d^{st} (-y_d^{pt} y_d^{si2} y_d^{i1r} + y_d^{pi2} y_d^{sr} y_d^{i1t}) L F_{2,1,1,-1}[m_q^P, m_q^s, \tilde{\mu}] + \\
& \quad \frac{1}{24} \tilde{\mu}^2 y_d^{pi2} \overline{y_d^{pr}} y_d^{i1r} (-5 g_1^2 + 3 g_2^2) L F_{2,1,1,0}[m_q^P, m_u^r, \tilde{\mu}] + \\
& \quad \frac{1}{8} \tilde{\mu}^2 y_d^{pi2} \overline{y_d^{pr}} y_d^{i1r} (g_1^2 + g_2^2) L F_{3,1,1,-1}[m_q^P, m_u^r, \tilde{\mu}] + \\
& \quad \frac{3}{4} \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{sr} y_d^{i112} \overline{y_d^{st}} y_d^{pt} L F_{2,2,1,-1}[m_q^P, m_u^t, m_q^s] + \\
& \quad \frac{1}{16} \tilde{\mu}^2 y_d^{pi2} \overline{y_d^{pr}} y_d^{i1r} (g_1^2 + g_2^2) L F_{2,2,1,-1}[m_q^P, \tilde{\mu}, m_u^r] - \\
& \quad \frac{3}{2} \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{sr} y_d^{i112} \overline{$$